A Model of the Gold Standard

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Historical Backdrop

- Gold standard dominated the **international monetary system** by the late 19th century
- Classical interpretation of the **Long Depression (1873–1896)**: gold supply could not accommodate rapid economic growth
- Mitchell (1975): **42% drop** in British wholesale prices 1873–1896
- Relief came only with **gold discoveries of the 1880s–90s** and the cyanidation process

Key Stylised Facts

- Inelastic gold supply $\Rightarrow$ deflation
- Cross-country shock transmission via specie flows
- Peripheral countries frequently imposed **bullion export controls**
- Inside money (bank deposits, notes) emerged to supplement gold

Framework: Lagos–Wright (2005)

- **Decentralized market (DM)** - buyer randomly matched with a seller with probability $\sigma \in (0,1)$, **seller produces divisible and perishable DM good** in quantity q from which only buyer derives utility, meetings are **anonymous** so payment must be immediate, gold as **means of payment**
- **Centralized market (CM)** - agents trade a **divisible and perishable CM good** where x is net consumption of CM good, any agent can produce CM good using linear technology

Preferences

Buyer (same across borders):

$$U(q,x) = u(q) + x$$

Seller in country i:

$$V^i(q,x) = -\omega^i q + x$$

Bellman equations I

At the beginning of period t buyer holds s units of money (i.e. gold or gold certificates). Gold entitles to a **dividend** δ and has a **price** ρ_t.
Buyer's Bellman equation is given by:

$$J(s,t) = \sigma[u(q_\omega(s,t)) + W(s - z_\omega(s,t),t)] + (1-\sigma)W(s,t)$$

CM value function:

$$W(s,t) = \max_{x,s'}[x + \beta J(s',t+1)]$$
$$\text{s.t.} \quad x + \rho_t s' = (\rho_t + \delta)s$$

$q_\omega(s,t)$ goods traded bilaterally for $z_\omega(s,t)$ units of money

Belman equations II

Seller in country $i \in \{a, b\}$ with productivity ω^i.
Seller's Bellman equation is given by:

$$K^i(s,t) = \sigma[-\omega^i q_{\omega^i}(\tilde{s},t) + B^i(s + z_{\omega^i}(\tilde{s},t),t)] + (1-\sigma)B^i(s,t)$$

CM value function:

$$B^i(s,t) = \max_{x,s'}[x + \beta K^i(s', t+1)]$$
$$\text{s.t.} \quad x + \rho_t s' = (\rho_t + \delta)s$$

where $\tilde{s}$ is money holdings of buyer

Bargaining

Terms of trade in DM market are determined by Nash bargaining:

$$\max_{q,z} [\underbrace{u(q) - (\rho_t + \delta)z}_{\text{(buyer)}}]^{\theta} [\underbrace{-\omega^i q + (\rho_t + \delta)z}_{\text{(seller)}}]^{1-\theta}$$

$$\text{s.t.} \quad z \leq s \quad \text{(liquidity constraint)}$$

$$-\omega^i q + (\rho_t + \delta)z \geq 0 \quad \text{(individual rationality)}$$

Buyer has all the bargaining power $\theta = 1$. The solution to this problem is:

$$q_\omega(s,t) = \begin{cases} \dfrac{(\rho_t + \delta)\, s}{\omega} & \text{if } s < \dfrac{\omega q^*(\omega)}{\rho_t + \delta} \\ q^*(\omega) & \text{otherwise} \end{cases}, \qquad z_\omega(s,t) = \begin{cases} s & \text{if } s < \dfrac{\omega q^*(\omega)}{\rho_t + \delta} \\ \dfrac{\omega q^*(\omega)}{\rho_t + \delta} & \text{otherwise.} \end{cases}$$

where $u'(q^*(\omega)) = \omega$ and $\frac{\omega}{\rho_t + \delta}$ is relative price of DM good

Equilibria

gold is scarce when price of gold is above fundamental price $\rho > \frac{\delta}{r}$

symmetric equilibrium

If $\omega^a = \omega^b = 1$, then:
$Q < (1-\beta)\frac{q^*}{\delta}$ and $\frac{\partial q}{\partial Q} > 0$
with $\rho > \delta/r$ and $\partial\rho/\partial Q < 0,\ \partial\rho/\partial\omega < 0,\ \partial q/\partial Q > 0,\ \partial q/\partial\omega < 0.$

Non-Neutrality of Money

Gold inflow ($Q\uparrow$) $\Rightarrow$ $\rho\downarrow$ but portfolio value $\uparrow\Rightarrow$ DM output $\uparrow$.

Supports **mercantilist** policies.

Secular Deflation

Productivity gain ($\omega\downarrow$) $\Rightarrow$ DM goods cheaper $\Rightarrow$ money demand $\uparrow\Rightarrow\rho\uparrow$.

Gold supply does not expand $\Rightarrow$ **productivity growth is deflationary**.

Asymmetric equilibrium

Permanent productivity improvement in country a only: $\omega^a = \omega < 1 = \omega^b$. **Specie flows**:

$$\Delta^a = \hat{Q}^a(\omega, Q) - Q > 0$$
$$\Delta^b = \underbrace{2Q - \hat{Q}^a(\omega, Q)}_{\text{final position}} - \underbrace{Q}_{\text{initial position}} < 0$$

Gold export controls in periphery.

Abundant gold stock

$$\rho(\omega, Q) = \frac{\delta}{r} \Leftrightarrow Q \geq \frac{r \max\{\omega q^*(\omega), q^*\}}{\delta(1+r)}$$

Inside Money & Banking I

Entrepreneurs with projects γ_j, distribution of which has support $[0, \tau], \tau > \beta^{-1}$. Funded by issuing debt claim in CM. Lenders entitled to receive $1 + \mu_t^i$ on fraction θ of project returns. Banks pool debt claims into deposits with return $\phi_t^i = \mu_t^i$.

- deposits are illiquid $\phi_t^i = r$, entrepreneurs with types $\gamma \in [\frac{1}{\beta}, \frac{1}{\beta\theta})$ are *credit constrained*
- deposits are liquid, then:

$$b_t^i = \eta \left(\tau - \frac{1 + \phi_t^i}{\theta} \right) \quad \text{(deposit market)}$$

$$1 + \phi_t^a = 1 + \phi_t^b = \frac{\delta + \rho_{t+1}}{\rho_t} \quad \text{(no arbitrage)}$$

$$1 + \phi = 1 + \frac{\delta}{\rho} \quad \text{(steady state)}$$

Inside Money & Banking II

With scarce aggregate liquidity:

$$\frac{\delta(1+r)Q}{r} + \underbrace{\eta(1+r)\left(\tau - \frac{1+r}{\theta}\right)}_{\text{Pareto improvement}} < q^*$$

Banking Crisis Transmission

A banking crisis in country a ($\theta^a \downarrow$):

- Deposit supply in a collapses
- Gold flows *to* a (rising ρ)
- ϕ falls equally in *both* countries (law of one price on gold)
- Country b: deposits $\uparrow$ but gold $\downarrow$ — net liquidity **falls**
- DM output **falls in** b despite no domestic shock